



Service Tool Instruction

Service Tool Instruction
Number: 3377783-3

Revision Level:

Released Date: 30-jul-2002

Description

Camshaft Gear Replacer Kit

Purpose

This document provides information for the use of the camshaft gear replacer kit, Part Number 3163054, on Cummins B3.9, B5.9, B5.9G, ISB, QSB5.9, C8.3, C8.3G, ISC, QSC8.3, ISL, and QSL9 series engines. The camshaft **must** be partially removed from the engine to use this tool.

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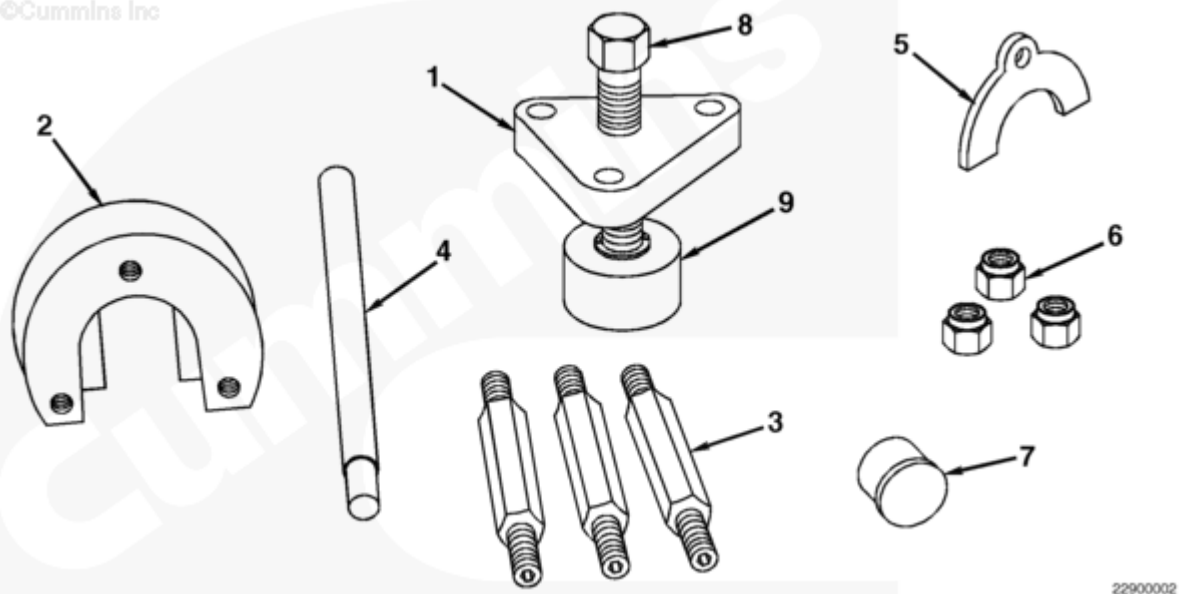


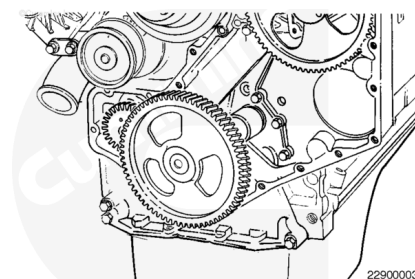
Table 1, Camshaft Gear Replacer Kit, Part Number 3163054

Item Number	Part Number	Description	Quantity
1	3164903	Replacer	1
2	3163055	Replacer plate	1
3	3164906	Stud	3
4	3164907	Handle	1
5	3824269	Retainer	1
6	3164753	Regular hexagon jam nut	3
7	3824098	Spacer	1

Table 1, Camshaft Gear Replacer Kit, Part Number 3163054			
Item Number	Part Number	Description	Quantity
8	3164902	Replacer screw	1
9	3164905	Replacer screw adapter	1
Not shown	3164904	Thrust bearing	1
Not shown	3164914	Roll pin	1

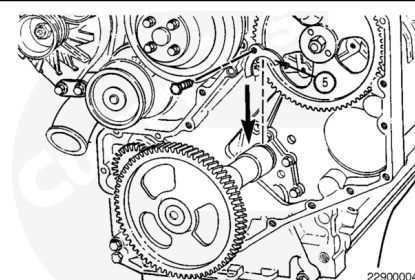
Table 2, Items Used with Camshaft Gear Replacer Kit, Purchased Separately			
Item Number	Part Number	Description	Quantity
Not shown	3822513	Tappet replacer kit (B3.9, B5.9, B5.9G, ISB, QSB5.9, C8.3, C8.3G, ISC, and QSC8.3 engines)	1
Not shown	3163468	Tappet replacer kit (ISL and QSL9 engines)	1

With the tappets held up using the appropriate tappet replacer, pull the camshaft out until the first two lobes are out of the block.



Note : On the 1991 model B5.9 engines equipped with an In-Line fuel pump, the camshaft gear mounting capscrew and washer must be removed.

Install the retainer (5) between the second camshaft lobe and second camshaft journal using an M8 x 1.25 capscrew secured in one of the camshaft thrust plate mounting holes. Tighten the capscrew.

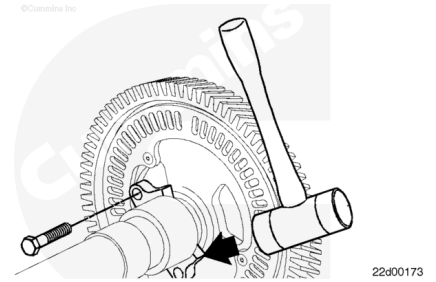


Note : On the ISC, QSC8.3, ISL, and QSL9 series engines, the thrust plate is captive and it will be

necessary to remove the thrust plate from the camshaft using a dead blow hammer and drift.

Do **not** reuse the thrust plate.

Temporarily install a 10-mm capscrew to prevent the thrust plate from rotating. Using a dead blow hammer and drift, strike the thrust plate to remove it from the camshaft.



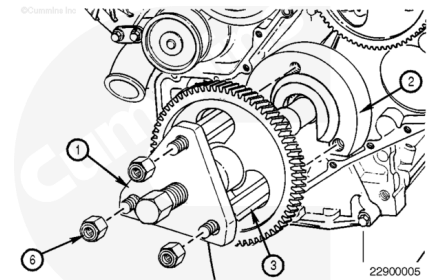
⚠ CAUTION ⚠

Make certain that the replacer plate is facing the correct way for gear removal. Failure to do so can result in damage to the tool and/or the camshaft.

Position the replacer plate (2) in the thrust plate groove.

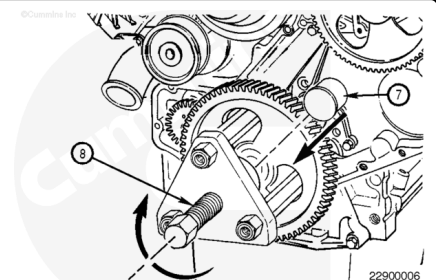
Install three studs (3) through the camshaft gear holes and into the replacer plate (2). Tighten the studs uniformly.

Install the replacer (1) onto the studs (3) using three regular hexagon jam nuts (6). Tighten the jam nuts uniformly.



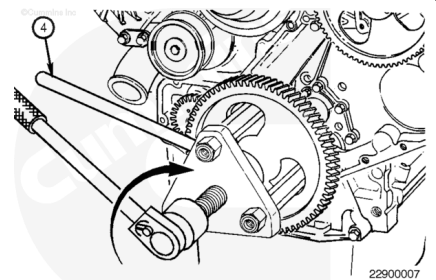
Position the spacer (7) between the replacer and the camshaft face.

Turn the replacer screw (8) **clockwise** by hand until it engages with the spacer (7) and camshaft face.



⚠ WARNING ⚠

The camshaft gear replacer can fall if not properly supported. To reduce the possibility of personal injury and component damage, use a hoist or other lifting device when using the camshaft gear replacer.



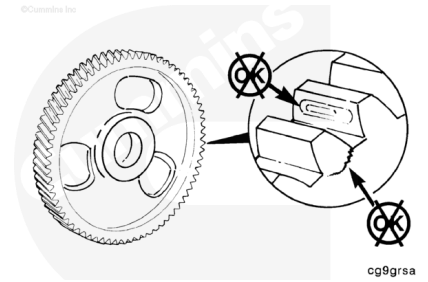
⚠ CAUTION ⚠

Do not use an impact wrench to turn the replacer screw on the replacer. Using an impact wrench can result in damage to the tool, camshaft gear, and camshaft.

Using the handle (4) to prevent the replacer from rotating, turn the replacer screw in the **clockwise** direction until the camshaft gear is removed from the camshaft.

Use a crocus cloth to remove any deposits from the camshaft and camshaft gear.

Inspect the camshaft and camshaft gear for nicks, burrs, and grooves.



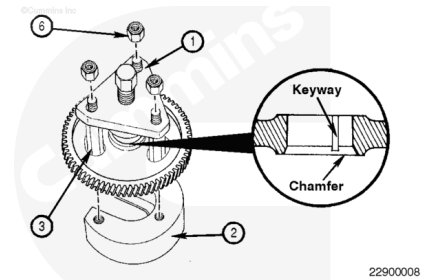
⚠ CAUTION ⚠

Make certain the replacer plate is positioned properly for installation. Failure to do so can result in damage to the tool, camshaft gear, or camshaft.

Position the camshaft gear so that the lead chamfer will be facing toward the block and the timing marks will be facing away from the block.

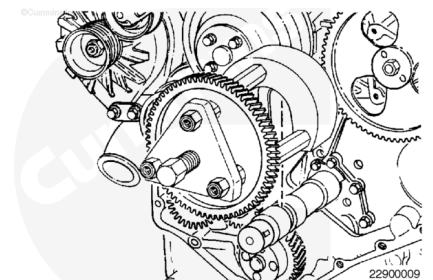
Install the three studs (3) through the camshaft gear holes and into the replacer plate (2). Tighten the studs uniformly.

With the replacer screw of the replacer (1) turned fully **counterclockwise**, install the replacer (1) on the studs (3) using three regular hexagon jam nuts (6). Tighten the jam nuts uniformly.



⚠ WARNING ⚠

The camshaft gear replacer can fall if not properly supported. To reduce the possibility of personal injury and component damage, use a hoist or other lifting device when using the camshaft gear replacer.



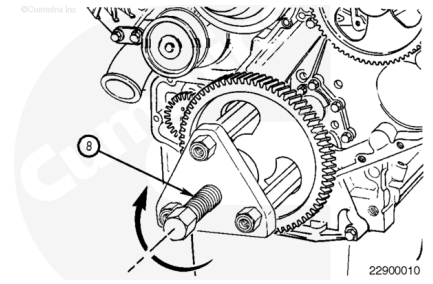
Note : On the ISC, QSC8.3, ISL, and QSL9 series engines, install a new thrust plate on the camshaft.

Lubricate the inside diameter of the camshaft gear and camshaft nose with clean engine oil.

Pick up the assembled tool and camshaft gear. Position the replacer plate behind the first journal of the camshaft.

Make sure the replacer, camshaft gear, and camshaft are properly aligned.

Turn the replacer screw (8) **clockwise** by hand until it engages with the camshaft gear.



⚠ CAUTION ⚠

Do not use an impact wrench to turn the screw on the screw press. Using an impact wrench can result in damage to the tool, camshaft gear, and camshaft.

Using the handle to prevent the tool from rotating, turn the replacer screw **clockwise** until the camshaft gear is positioned against the camshaft shoulder (1).

Remove the camshaft gear replacer kit, Part Number 3163054, and the tappet replacer from the engine.

